

Math1b Parameterisation

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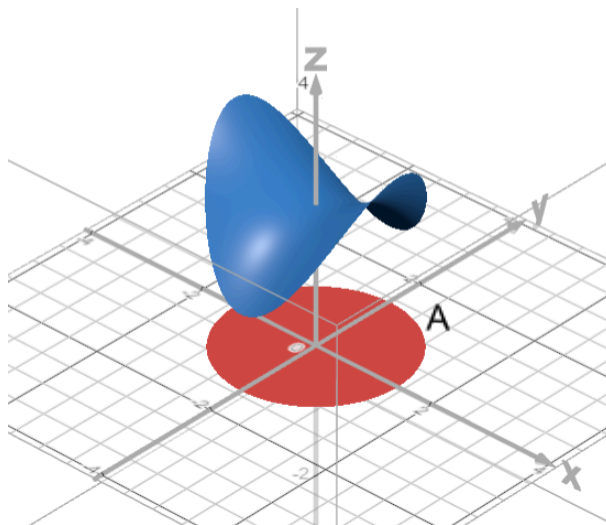
Basics:

What will you need parameterisation for?

- Calculating the value of a function integrated over a specific domain
- Calculating length, area or volume of a set
- Calculating the work done by a vector field
- Doing a boundary investigation when finding extrema

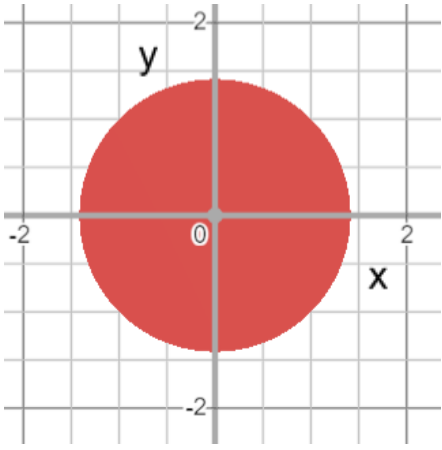
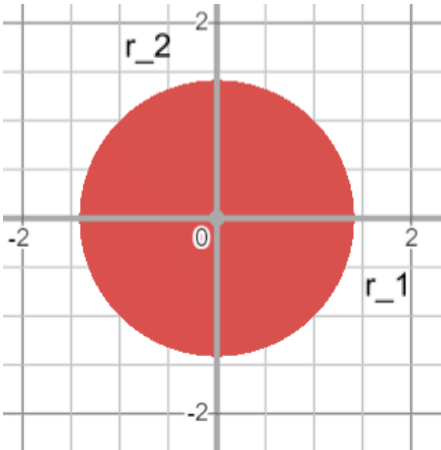
What is parameterisation and why do we do it?

We use parameterisation for many things, but it is always about making a set easier to interpret and calculate with. One example is when it is not obvious what the input values of our function should be, and more generally, when it is difficult to otherwise describe the set we are working with. In the case below we have a function of 2 variables restricted to a domain which is not easily defined, in terms of each of the variables. Let's call the circle A



The diagram displays a blue function $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2} + 2.5$ restricted to the domain of the red circle ie. $f: A \rightarrow \mathbb{R}$. When parameterising for a function, we only work with the domain, and then restrict the function to that parameterised domain.

In the diagrams below, we can see how parameterisation makes it easier to work with sets in certain situations. Specifically when we want an easy way to find points within a set. With parameterisation, each point is described by a general vector function r , where we know all the input values.

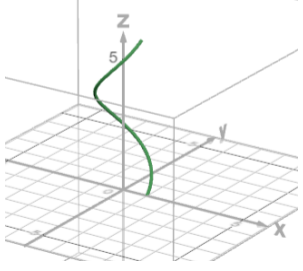
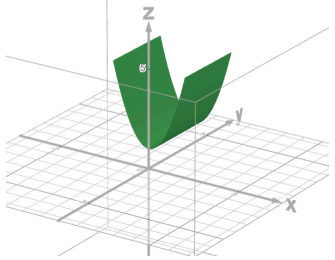
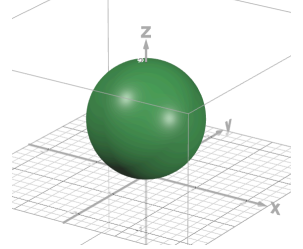
$x^2 + y^2 < \sqrt{2}^2$ $A = \{ (x, y) \mid x^2 + y^2 < \sqrt{2}^2, x, y \in \mathbb{R} \}$	$r(u, v) = [\sqrt{2} \cdot v \cdot \cos(u), \sqrt{2} \cdot v \cdot \sin(u)]$ $= [r_1(u, v), r_2(u, v)]$ $u \in [0, 2\pi]$ $v \in [0, 1]$ $\Gamma = [0, 2\pi] \times [0, 1]$ $A = \{ (r_1(u, v), r_2(u, v)) \mid u \in [0, 2\pi], v \in [0, 1] \}$
	

On the left we have the set A described using a relation between x and y, but we have no way of figuring out which x and y values uphold this relation other than just testing values. On the other hand, if we parameterise A, we get a formula which exactly describes what the two coordinates need to be (They are in terms of u and v which are both known), making restricting the function really simple. The two sets are identical, so we would get exactly the same values but in a more manageable format. That is the main point.

What situations can we be in?

Our domain can come in many forms, but we will likely only work with cases in 1D, 2D or 3D. No matter what the shape ends up being, it can always be described by exactly as many parameters as its dimension. For example, a 2D surface can always be described by 2 parameters in the parameterisation. In the diagrams below, all shapes exist in 3D space, but their geometries are different.

Case	Parameterisation (Γ is always axis-parallel)
1D Geometry - curves and lines	$r(u) = [\cos(u), \sin(u), u]$ $u \in [0, 6] = \Gamma$

	(forms a line)
2D Geometry - surfaces 	$r(u, v) = [u, v, 2u^2 + 1]$ $u \in [-1, 1]$ $v \in [0, 2]$ $\Gamma = [-1, 1] \times [0, 2]$ (forms a rectangle)
3D Geometry - volumes 	$r(u, v, w) = [w \cos(v) \sin(u), w \sin(u) \sin(v), w \cos(u) + 1]$ $u \in [0, \pi]$ $v \in [0, 2\pi]$ $w \in [0, 1]$ $\Gamma = [0, \pi] \times [0, 2\pi] \times [0, 1]$ (forms a cuboid)

What are the most likely sets you'll see:

- Ellipses
- General functions
- Straight lines

Formulas:

Ellipses

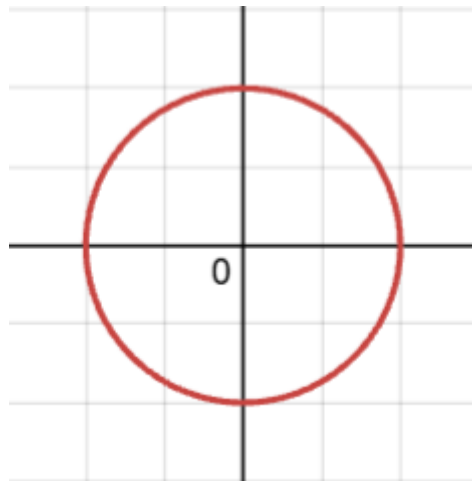
To start off we make a circle. To make a circle of radius 1 we use:

$$r(u) = [\cos(u), \sin(u)]$$

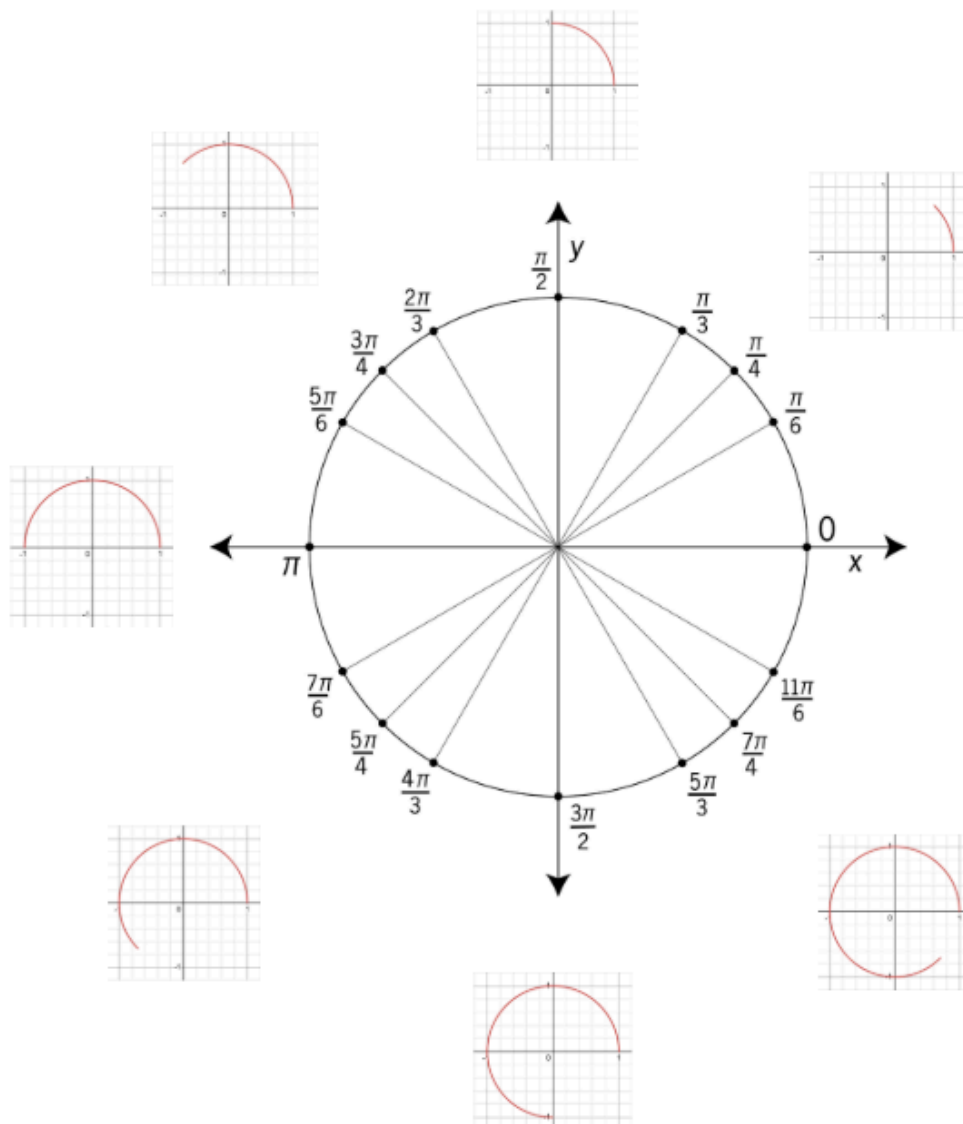
$$u \in [0, 2\pi]$$

This is our starting point. There are 3 types of changes we can do to this parameterisation

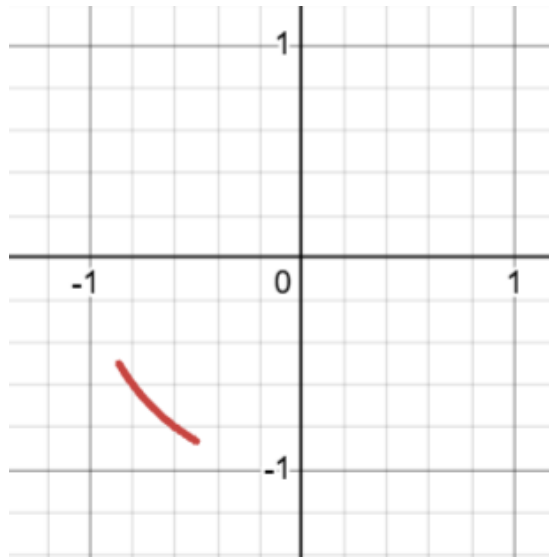
1. Change the arc length
2. Move the origin
3. Morph the x and y radii



To get arcs of a circle, you simply have to pick the two angles you want your arc to be between. The easiest way is to look at the unit circle. All the examples are from 0 to some angle, but you can pick any two and imagine what an arc between them would look like. You then simply have to set you $u \in [\theta_1, \theta_2]$ where $\theta_1 < \theta_2$



Here is an example of a piece which is $u \in [\frac{7\pi}{6}, \frac{4\pi}{3}]$.



Our form now looks like this:

$$r(u) = [\cos(u), \sin(u)]$$

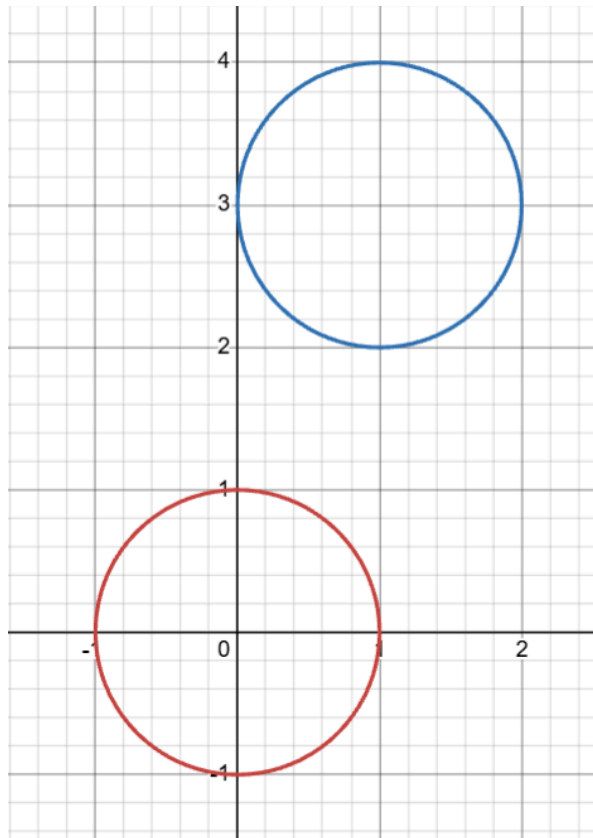
$$u \in [\theta_1, \theta_2]$$

By default, all parameterisations will have the center (0,0). The point might not always be included, but it will be quite clear that it is based around it:

Ex

To move it you simply have to add to the x or y coordinate. Let's say you want it to center around (c_1, c_2)

$$r = [r_1 + c_1, r_2 + c_2]$$



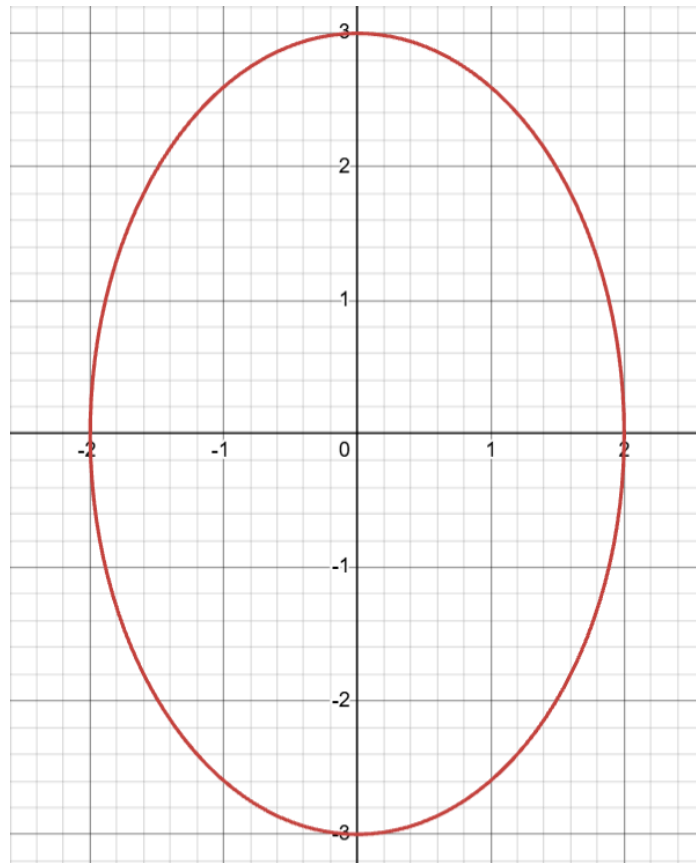
Here we have moved the unit circle to (1,2). $r(u) = [\cos(u) + 1, \sin(u) + 2]$

Our form now looks like this:

$$r(u) = [\cos(u) + c_1, \sin(u) + c_2]$$

$$u \in [\theta_1, \theta_2]$$

We can change the circle into an ellipse by multiplying by the semi-axis on each of the coordinates. Since we are working with coordinates, it will simply scale them linearly by the multiplied factor. Here the new “x-radius” is 2 and the “y-radius” is 3. In general we call the x-semi axis a and the y semi-axis b . If both semi-axis are the same $a = b$, you will simply increase the radius of your original circle.



$$r(u) = [2\cos(u), 3\sin(t)]$$

Our form now looks like this:

$$r(u) = [a \cdot \cos(u) + c_1, b \cdot \sin(u) + c_2]$$

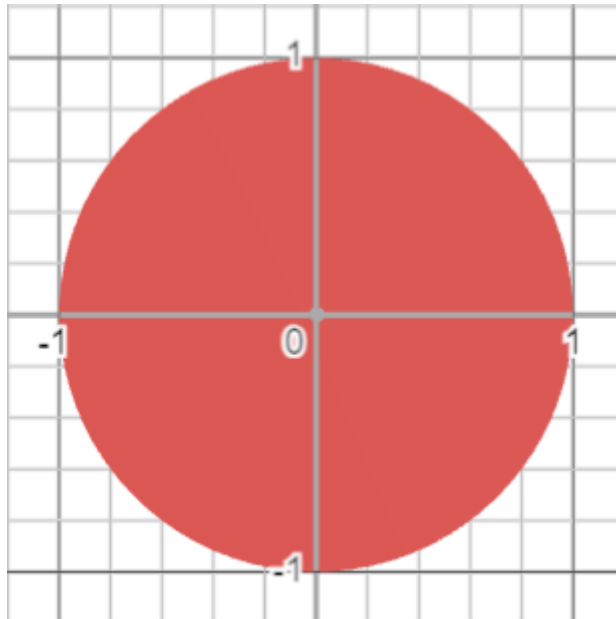
$$u \in [\theta_1, \theta_2]$$

We have now seen all you can do to an ellipse in 1D geometry. If we want to level up to a disk, we have to add another parameter as we will enter 2D geometry. All previous operations changes still hold (Arclength, origin translation and radii-morphing) but now with an extra dimension we gain an extra option: Inner and outer radius. The form of a unit disk is as follows:

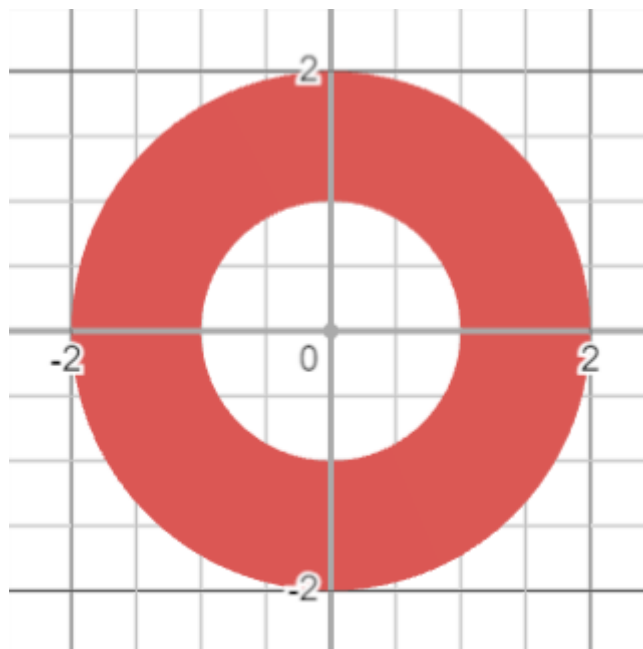
$$r(u, v) = [v \cdot \cos(u), v \cdot \sin(u)]$$

$$u \in [0, 2\pi]$$

$$v \in [0, 1]$$



If you are working with a circular disk, ie $a = b = r_c$ (*radius of circle*) for your semi-axis, it is quite straight forward. If you want your inner radius to be r_{inner} then you simply have to replace r_c with $v \in [r_{inner}, r_c]$. Here is an example where the radius of 2 and and inner radius of 1:



$$\begin{aligned} r(u, v) &= [v \cdot \cos(u), v \cdot \sin(u)] \\ u &\in [0, 2\pi] \\ v &\in [1, 2] \end{aligned}$$

It is unlikely you will have to do it for an ellipse.

In the end, we can combine everything into 2 formulas. One for a circular disk and another for an elliptical disk

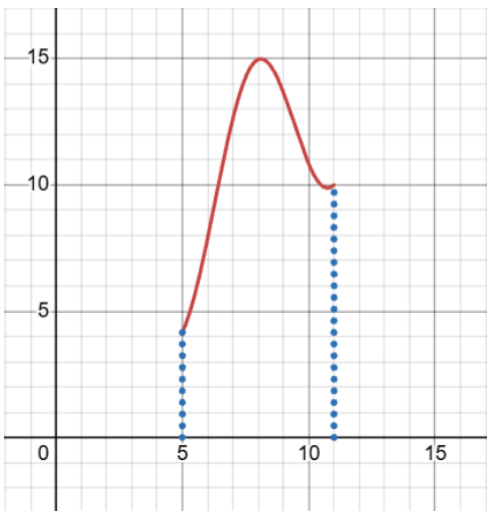
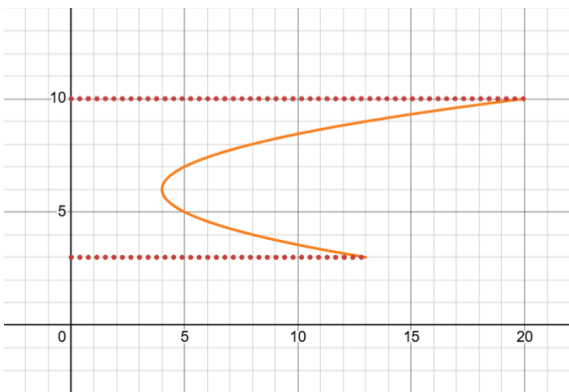
Circular disk	Elliptical disk
$r(u, v) = [v \cdot \cos(u) + c_1, v \cdot \sin(u) + c_2]$ $u \in [\theta_1, \theta_2]$ $v \in [r_{inner}, r_c]$ For full disk $r_{inner} = 0$	$r(u, v) = [a \cdot v \cdot \cos(u) + c_1, b \cdot v \cdot \sin(u)]$ $u \in [\theta_1, \theta_2]$ $v \in [0, 1]$

You can test out the parameterisations in desmos here:

[Desmos, circular disk](#)

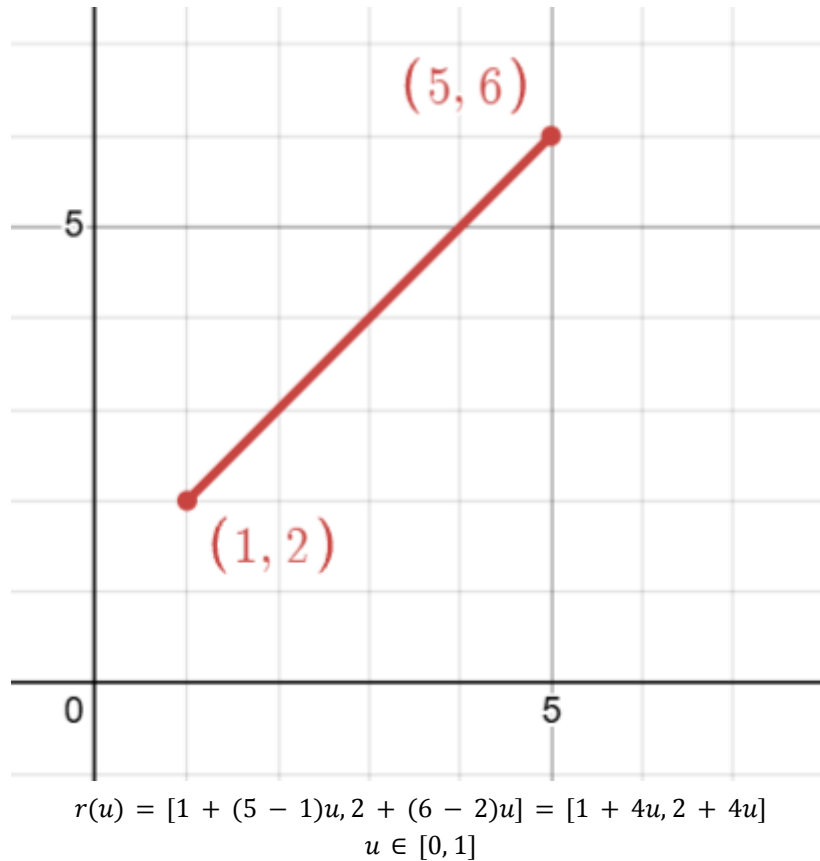
[Desmos, elliptical disk](#)

Parameterisation of general functions:

$g(x) = 4\sin(x) + 3 + x$ $x_1 = 5 < x < x_2 = 11$ $r(u) = [u, g(u)]$ $u \in [x_1, x_2]$	$h(y) = (y - 6)^2 + 4$ $y_1 = 3 < y < y_2 = 10$ $r(u) = [h(u), u]$ $u \in [y_1, y_2]$
	

When we are given any function like this, the simplest way to parameterise them is just by replacing the coordinates by the function values.

Any straight line between any two points x_0, x_1



Let's say you are given the following points:

$$x_0 = [a_1, a_2, \dots, a_n]$$

$$x_1 = [b_1, b_2, \dots, b_n]$$

You essentially start at point x_0 and add the vector direction from x_0 to x_1

$$x(t) = x_0 + (x_1 - x_0)t, t \in [0, 1]$$

$$r(u) = [a_1 + u(b_1 - a_1), a_2 + u(b_2 - a_2), \dots, a_n + u(b_n - a_n)], u \in [0, 1]$$

If $x_0 = 0$ ie the line starts at the origin, this simplifies to:

$$r(u) = [ub_1, ub_2, \dots, ub_n], u \in [0, 1]$$

Extra trick:

If you ever need to reverse the direction of your parameterisation, it is actually very simple. Remember that the drawing direction of your parameterisation is dependent on the tangent vector $r'(u)$. So if you want to reverse it, you just have to find a parameterisation that does exactly the same thing, but gives you the points in a reverse order with a negative tangent vector. That is given by $t(u)$

$$r(u), u \in [a, b]$$

$$r(u) \rightarrow t(u) = r(a + b - u)$$

$$t'(u) = -r'(a + b - u)$$

$$t(a) = r(b)$$

$$t(b) = r(a)$$